

CodioProjectFileEditFindViewToolsEducationHelpConfigure...Virtual DesktopConfigure...

Terminal

exercise4.py

xlibrary.py

book.py

1from library import Library

2from book import Book

3

4library = Library()

5book1 = Book('Three Musketeers', 'Alexandre Dumas', 'fiction')

6book2 = Book('The Count of Monte Cristo', 'Alexandre Dumas', 'fiction')

7book3 = Book('Educated', 'Tara Westover', 'nonfiction')

8

9library.add_book(book1)

10library.add_book(book2)

11library.add_book(book3)

12library.sort_books()

13

14print(library.books)

15print(library.fiction)

16print(library.nonfiction)

17print(library.search_author('Alexandre Dumas'))

18print(library.search_author('Herman Melville'))

19print(library.search_title('Educated'))

20print(library.search_title('Moby Dick'))

Guide

x

Collapse

Advanced Topics -> Coding Exercises - Advanced Topics

Advanced Topics Exercise 4

Exercise 4

You are given code for the `Library` class (in its own file). This is a composite class. You are going to create the `Book` class (the component class) in `book.py` file. Look over the `Library` class carefully to determine what attributes are needed for the `Book` class. In addition, the table of output contains a hint as to what method the `Book` class needs.

```
library = Library()
book1 = Book('Three Musketeers', 'Alexandre Dumas', 'fiction')
book2 = Book('The Count of Monte Cristo', 'Alexandre Dumas', 'fiction')
book3 = Book('Educated', 'Tara Westover', 'nonfiction')

library.add_book(book1)
library.add_book(book2)
library.add_book(book3)
library.sort_books()
```

The following commands should produce the output on the right.

Command	Output
<code>print(library.books)</code>	[Book(Three Musketeers, Alexandre Dumas, fiction), Book(The Count of Monte Cristo, Alexandre Dumas, fiction), Book(Educated, Tara Westover, nonfiction)]
<code>print(library.fiction)</code>	[Book(Three Musketeers, Alexandre Dumas, fiction), Book(The Count of Monte Cristo, Alexandre Dumas, fiction)]
<code>print(library.nonfiction)</code>	[Book(Educated, Tara Westover, nonfiction)]
<code>print(library.search_author('Alexandre Dumas'))</code>	[Book(Three Musketeers, Alexandre Dumas, fiction), Book(The Count of Monte Cristo, Alexandre Dumas, fiction)]
<code>print(library.search_author('Herman Melville'))</code>	[]
<code>print(library.search_title('Educated'))</code>	True
<code>print(library.search_title('Moby Dick'))</code>	False

TRY IT

```
[Book(Three Musketeers, Alexandre Dumas, fiction),
Book(The Count of Monte Cristo, Alexandre Dumas,
fiction), Book(Educated, Tara Westover, nonfiction)]
[Book(Three Musketeers, Alexandre Dumas, fiction),
Book(The Count of Monte Cristo, Alexandre Dumas,
fiction)]
[Book(Educated, Tara Westover, nonfiction)]
[Book(Three Musketeers, Alexandre Dumas, fiction),
Book(The Count of Monte Cristo, Alexandre Dumas,
fiction)]
[]
True
False
```

Submit your code for evaluation

Here is one potential solution:

```
class Book:
    def __init__(self, title, author, genre):
        self.author = author
        self.title = title
        self.genre = genre

    def __repr__(self):
        return f'Book({self.title}, {self.author}, {self.genre})'
```

In the `library.py` file, you can see references to the three attributes needed for the `Book` class. The attribute `title` is referenced on line 15, the attribute `author` is referenced on line 23, and the attribute `genre` is referenced on lines 36 and 44.

When you print a list and you want to a string representation of a user-defined object, you need to override the `__repr__` method.

Check !!!

LAST RUN on 12/28/2023, 3:22:56 AM

Test passed

Next

95% (19:41)

https://codio.com/sandipan-dey/coding-exercises-9/tree/code%2Fadvanced%2Fexercise4.py

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